

Mild Cognitive Impairment (MCI)

Mild cognitive impairment (MCI) is an early stage of memory loss or other cognitive ability loss (such as language or visual/spatial perception) in individuals who maintain the ability to independently perform most activities of daily living.

MCI describes cognitive changes that are serious enough to be noticed by the person affected and by family members and friends but do not impact the individual's ability to carry out everyday activities. Studies estimate that approximately 12-18% of people age 60 or older are living with MCI.

MCI can develop for multiple reasons, and individuals living with MCI may or may not go on to develop dementia. For neurodegenerative diseases, MCI can be an early stage of the disease continuum, including for Alzheimer's disease, if the hallmark changes in the brain are present.

Data from at least one community-based study suggests that, among those with MCI, approximately 15% develop Alzheimer's or another dementia after two years. In some individuals, however, MCI reverts to normal cognition or remains stable. In other cases, a medication or non-dementia disease may cause cognitive impairment in the absence of MCI, and symptoms may be reversed if the medication is ceased. It is important that people experiencing cognitive changes seek help as soon as possible for diagnosis and possible treatment.

Symptoms

Experts classify mild cognitive impairment based on the thinking skills affected:

- **Amnesic MCI:** MCI that primarily affects memory. A person may start to forget important information that he or she would previously have recalled easily, such as appointments, conversations or recent events.
- **Nonamnesic MCI:** MCI that affects thinking skills other than memory, including the ability to make sound decisions, judge the time or sequence of steps needed to complete a complex task, or visual perception.

Diagnosis

Mild cognitive impairment is a clinical diagnosis representing a doctor's best professional judgment about the reason for a person's symptoms. Individuals living with MCI who have an abnormal brain positron emission tomography (PET) scan or spinal

fluid test for amyloid beta protein, which is the protein in amyloid plaques (one of the two hallmarks of Alzheimer's), are considered to have a diagnosis of MCI due to Alzheimer's disease.

The Alzheimer's Association® partnered with the National Institute on Aging (NIA) to convene expert workgroups to update the diagnostic guidelines for MCI due to Alzheimer's disease in 2024, suggesting that, in some cases, MCI is an early stage of Alzheimer's or another dementia.

The guidelines recommend finding a biomarker (a measurable biological factor, such as levels of a protein, that indicates the presence or absence of a disease) for people with MCI to learn whether they have brain changes that put them at high risk of developing Alzheimer's and other dementias.

If it can be shown that changes in the brain, cerebrospinal fluid and/or blood are caused by physiologic processes associated with Alzheimer's, the revised guidelines recommend a diagnosis of MCI due to Alzheimer's disease.

A medical workup for MCI includes the following core elements:

- Thorough medical history, where the physician documents current symptoms, previous illnesses and medical conditions, and any family history of significant memory problems or dementia.
- Assessment of independent function and daily activities, which focuses on any changes from a person's usual level of function.
- Input from a family member or trusted friend to provide additional perspective on how function may have changed.
- Assessment of mental status using brief tests designed to evaluate memory, planning, judgment, ability to understand visual information and other key thinking skills.
- In-office neurological examination to assess the function of nerves and reflexes, movement, coordination, balance and senses.
- Evaluation of mood to detect depression; symptoms may include problems with memory or feeling "foggy." Depression is widespread and may be especially common in older adults.
- Laboratory tests including blood tests and imaging of the brain's structure.

If the workup doesn't create a clear clinical picture, the doctor may recommend neuropsychological testing, which involves a series of written or computerized tests to evaluate specific thinking skills.

See alz.org/diagnosticcriteria for more information.

Causes and risks

The causes of MCI are not yet completely understood. Experts believe that many cases — but not all — result from brain changes occurring in the very early stages of Alzheimer's or other neurodegenerative diseases that cause dementia.

The risk factors most strongly linked to MCI when the underlying cause is neurodegenerative disease and not another cause are advancing age, family history of Alzheimer's or another dementia, and conditions that raise risk for cardiovascular disease.

Treatment and outcomes

Donanemab (Kisunla™) and lecanemab (Leqembi®) have received traditional approval by the U.S. Food and Drug Administration for the treatment of early Alzheimer's disease

These therapies address the underlying biology of Alzheimer's and slow disease progression. They demonstrate that removing beta-amyloid plaques, one of the hallmarks of Alzheimer's, from the brain slows clinical decline with benefits to both cognition and function in people living with the disease.

The treatment landscape is rapidly changing. For the most up-to-date information on FDA-approved treatments for Alzheimer's disease, visit alz.org/medications.

When considering any treatment, it is important to have a conversation with a health care professional to determine whether it is appropriate. A clinician who is experienced in using these types of medications should monitor people who are taking them and ensure that the recommended guidelines are strictly observed.

More research is needed on the biological changes associated with normal aging, MCI, and Alzheimer's and other dementias to better understand the causes of and risk factors for MCI and the prognosis for those with the condition.

Individuals who have been diagnosed with MCI should be reevaluated every six months to determine if symptoms have progressed.

Visit **alz.org/mci** to learn more.

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